

Component Name	Purpose / Specification
MG996R Metal-Gear Servo	High-torque 180° servo; 3 per leg × 6 legs, plus 2 spares
ESP32 Dev Board	Main microcontroller with Wi-Fi/Bluetooth
PCA9685 16-Channel Servo Driver	External PWM control for servos
ESP32-CAM Camera Board	Remote visual observation
HC-SR04 Ultrasonic Distance Sensor	Short-range obstacle detection
MPU6050 IMU	Acceleration and gyroscope measurements
VL53L0X Time-of-Flight Distance Sensor	More precise laser/ToF distance sensing
18650 Li-ion Battery Cell	Robot power source
4-Cell 18650 Battery Holder	Holds four 18650 cells
4S 40A Li-ion BMS / Protection Board	Battery protection and management
LM2596 Buck Regulator	Voltage regulation for electronics
PLA Filament 1.75 mm, 1 kg	3D-printed body and leg components
26 AWG Hook-up Wire Kit	Internal electrical wiring
Jumper Wire Kit	Prototype and controller connections
TOTAL ESTIMATED BOM COST	

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Quantity	Unit Cost Range (ZAR)	Minimum Cost (ZAR)	Maximum Cost (ZAR)
20	R85-R144	R 85.00	R 144.00
1	R98	R 98.00	R 98.00
2	R98	R 98.00	R 98.00
1	R184-R349	R 184.00	R 349.00
2	R24	R 24.00	R 24.00
1	R86.80-R170	R 86.80	R 170.00
1	R139-R295	R 139.00	R 295.00
4	R95-R110	R 95.00	R 110.00
1	R18-R28	R 18.00	R 28.00
1	R43	R 43.00	R 43.00
1	R16-R28	R 16.00	R 28.00
1	R229-R369	R 229.00	R 369.00
1	R58-R69	R 58.00	R 69.00
1	R15	R 15.00	R 15.00
		R 1,188.80	R 1,840.00

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Supplier Name	Supplier Stock Code
Communica / Make Electronics	CMU SERVO MOTOR MG996R 11KG 180D / 9ME4004
Micro Robotics	ESP32-DEV
Micro Robotics	PCA9685
Neotronics / PriceCheck market listing	ESP32-CAM / market listing
Micro Robotics	HC-SR04
Mantech / PriceCheck market listing	15M8258 / MPU6050
Micro Robotics	VL53LOX-SEN / 2490
Communica	LC18650-TAG / INR18650-25HW
Micro Robotics	18650-WIRE-4X / 18650-PCB-4X
Communica	BMT 4S LITH BATT CHARGE/PROT 40A
Micro Robotics	LM2596-MINI / LM2596-3POT
Micro Robotics / iStore	SAF-PLA-102 / SMD-3301010061
Micro Robotics	WIRE40 / WIRE-KIT
Micro Robotics	JUMP-MM-L10

Direct URL / Link to Buy

<https://www.communica.co.za/products/cmu-servo-motor-mg996r-11kg-180d> ;

<https://make.net.za/product/mg996r-servo-motor-analog-180-degrees/>

<https://www.robotics.org.za/development-boards/ESP32-DEV>

<https://www.robotics.org.za/PCA9685>

https://neotronics.co.za/index.php?product_id=1043&route=product/product ;

<https://www.pricecheck.co.za/offers/209879132/ESP32->

[CAM%2BDevelopment%2BBoard%2BWifi%2B%2BBluetooth%2B%2B2MP%2BCamera](https://www.pricecheck.co.za/offers/209879132/ESP32-CAM%2BDevelopment%2BBoard%2BWifi%2B%2BBluetooth%2B%2B2MP%2BCamera)

<https://www.robotics.org.za/HC-SR04>

<https://www.mantech.co.za/Stock.aspx?Query=Gyroscopeand> ;

<https://www.pricecheck.co.za/offers/248867635/Accelerometer%2BModule%2B-%2BMPU6050>

<https://www.robotics.org.za/sensors/time-of-flight-sensors>

<https://www.communica.co.za/products/lc18650-tag> ;

<https://www.communica.co.za/products/inr18650-25hw>

<https://www.robotics.org.za/18650-WIRE-4X> ;

<https://www.robotics.org.za/18650-PCB-4X>

<https://www.communica.co.za/products/bmt-4s-lith-batt-charge-prot-40a>

<https://www.robotics.org.za/LM2596-MINI> ; <https://www.robotics.org.za/LM2596-3POT>

<https://www.robotics.org.za/SAF-PLA-102> ; <https://www.istore.co.za/creality-cr-pla-1kg-filament-black-smd-3301010061>

<https://www.robotics.org.za/WIRE40> ; <https://www.robotics.org.za/WIRE-KIT>

<https://www.robotics.org.za/JUMP-MM-L10>

Total Cost Range (ZAR)	Notes
R 1,700.00	18 required for the robot; 2 spares recommended.
R 98.00	Current listed price includes VAT.
R 196.00	Two boards provide up to 32 servo channels.
R 184.00	Price depends on exact ESP32-CAM version/supplier.
R 48.00	2-400 cm nominal range.
R 86.80	Use one 6-axis IMU for orientation/motion estimation.
R 139.00	Choose one model; the price range reflects two current Micro Robotics listings.
R 380.00	Exact battery configuration must be confirmed with the servo power design.
R 18.00	Select wire or PCB holder to match the final wiring.
R 43.00	Only use if the final battery pack is designed as a 4S pack.
R 16.00	Select the regulator according to measured load/current.
R 229.00	One 1 kg spool is a starting allowance; actual usage depends on CAD design.
R 58.00	Includes multiple colours for power/signal identification.
R 15.00	20-piece male-to-male jumper set.
R 1,188.80 – R 1,840.00	





AEGIS

HEXAPOD ROBOT

INNOVATION | MOBILITY | SAFER TOMORROWS



MARKET
REAL DEMAND



ECONOMIC VALUE
JOBS, GROWTH, INNOVATION



SOUTH AFRICA & AFRICA
LOCAL IMPACT, LASTING CHANGE

TIONS